SOFTWARE QUALITY ASSURANCE & VERIFICATION SPECIFICATION

HELPBOT: QUALITY & VERIFICATION EVIDENCE

IEEE 829 Test Suite, Boundary Analysis, Decision Logic, Defect Log & Quality Metrics

1. Document Control & Quality Governance

Project Reference	Case Study No. 97 — Regional Broadband Service Provider	**Test Standard**	IEEE Std 829-2008 Verification Format
Traceability Baseline	100% Coverage (FR-01 to FR-10, NFR-01 to NFR-07)	**Target Ingress**	Web Self-Care & WhatsApp Business Cloud API
Exit Gate Policy	Zero Critical or High severity defects; Defect Removal Efficiency (DRE) ≥ 80%; 100% pass on Handover Decision Rules.		

2. Master Verification Suite: Functional & Quality Test Cases

Test ID	Traceability	Test Condition & Input Scenario	Expected Observable Outcome	Status
TC-01	FR-01 (Ingress)	Submit concurrent query via Web widget and WhatsApp API.	Both sessions ingest payload; context preserved with 0 dropouts.	PASS
TC-02	FR-02 (Auth)	Submit valid Registered Mobile Number; enter 6-digit OTP.	Identity authenticated; session token issued with 15-min TTL.	PASS
TC-03	FR-03 (Bill Query)	Authenticated user requests current invoice and balance due.	Returns balance, due date, and authenticated PDF download URL.	PASS
TC-04	FR-04 (Plan Change)	Select 100 Mbps to 300 Mbps upgrade; confirm via OTP.	Billing database updates tariff profile; sends WhatsApp receipt.	PASS
TC-05	FR-05 (Diagnostics)	Trigger slow broadband diagnostic workflow on active line.	OLT queried; returns optical Rx (-19.2 dBm) and sync confirmation.	PASS
TC-06	FR-06 (Outage)	Submit location query during simulated node maintenance.	Identifies active NOC incident; displays estimated restoration time.	PASS
TC-07	FR-07 (Handover)	NLU classification returns confidence score C = 0.55 (< 0.70).	Bot halts automated intent; routes session to agent queue.	PASS
TC-08	FR-08 (Context)	Execute handover escalation to agent CRM desktop.	Agent console receives full chat history and customer ID in ≤ 3.0s.	PASS
TC-09	FR-09 (Payments)	Generate dynamic UPI link for outstanding bill amount.	Tokenized link generated; webhook receives instant confirmation.	PASS
TC-10	FR-10 (Analytics)	Complete resolution flow; query deflection warehouse.	Record logged with resolution flag, intent ID, and latency metric.	PASS
TC-11	NFR-01 (Accuracy)	Run 500-phrase evaluation benchmark against the 5 domains.	Intent classification precision reaches 93.4% (≥ 92.0% target).	PASS
TC-12	NFR-02 (Latency)	Inject 1,000 synthetic queries under sustained peak load.	Chatbot response time p95 measures 1.42s (target ≤ 1.80s).	PASS
TC-13	NFR-04 (Privacy)	Inspect network trace and logs during card/OTP submission.	Zero plaintext credentials or PAN data recorded; TLS 1.3 enforced.	PASS
TC-14	NFR-06 (Coupling)	Simulate 5-second outage / timeout on Billing API endpoint.	Resilience4j opens; bot gracefully offers deferred callback.	PASS

3. Boundary Value Analysis (BVA) & Equivalence Partitioning

BVA targets sensitive operational boundaries around the **confidence score threshold (C = 0.70)**, handover response latency, and failed-query retry limits.

3.1 Boundary Value Analysis: Intent Confidence Score (Threshold = 0.70)

Test Case ID	Input Confidence (C)	Boundary Classification	Observed System Behavior	Result
BVA-01	C = 0.68	Interior Below (Invalid)	Classified uncertain; triggers human handover routine	PASS
BVA-02	C = 0.69	Just Below Boundary	Handover initiated; transcript packaged to agent queue	PASS
BVA-03	C = 0.70	Exact Critical Threshold	Intent accepted; proceeds with automated resolution	PASS
BVA-04	C = 0.71	Just Above Boundary	Automated action executed without human intervention	PASS
BVA-05	C = 0.95	Interior Above (Valid)	Automated intent executed; instant user response rendered	PASS

3.2 Equivalence Partitioning Summary

Partition Domain	Valid Partition (Automated Path)	Borderline Partition (Confirmation)	Invalid Partition (Handover Path)
Confidence Score (C)	0.70 ≤ C ≤ 1.00 (Auto-Resolve)	0.60 ≤ C < 0.70 (Clarification Prompt)	0.00 ≤ C < 0.60 (Direct Escalation)
Query Classification	5 Core Drivers (Bill, Plan, Speed, Outage, Pay)	Ambiguous multi-intent inquiry	Out-of-scope (Hardware, Legal, Harassment)
Retry Counter	Attempt = 1 (Normal processing)	Attempt = 2 (Prompt reformulation)	Attempt ≥ 3 (Immediate Handover)

4. Handover Decision Table (Multi-Factor Escalation Rules)

Rule Conditions & Triggers	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5	Rule 6
Intent Confidence Score (C ≥ 0.70)?	Yes	No	Yes	Yes	No	Irrelevant
Query in Top 5 In-Scope Categories?	Yes	Yes	No	Yes	No	Irrelevant
User Explicitly Requested Human Agent?	No	No	No	Yes	No	Yes
Consecutive Unresolved Attempts ≥ 2?	No	No	No	No	Yes	Irrelevant
Backend API Available (Circuit Closed)?	Yes	Yes	Yes	Yes	Yes	No (Open)
System Action Executed						
Execute Automated Self-Service Flow	**X**	—	—	—	—	—
Initiate Handover to Live Support Agent	—	**X**	**X**	**X**	**X**	**X**
Push Context & Transcript in ≤ 3.0s	—	**X**	**X**	**X**	**X**	**X**
Verification Outcome	PASS	PASS	PASS	PASS	PASS	PASS

5. Empirical Defect Discovery & Resolution Log

Bug ID	Severity	Root Cause & Defect Description	Phase Detected	Resolution	Status
BUG-01	CRITICAL	Handover router dropped conversation state during WhatsApp webhook reconnection spikes.	Integration QA	Persisted chat state in Redis sentinel cluster.	CLOSED
BUG-02	HIGH	Confidence score of 0.698 rounded up to 0.70, incorrectly preventing human handover.	Integration QA	Enforced strict floating-point ceiling comparisons.	CLOSED
BUG-03	HIGH	Billing adapter blocked conversational thread when legacy server timed out (> 5.0s).	Adapter Testing	Wrapped call in Resilience4j circuit breaker.	CLOSED
BUG-04	MEDIUM	Regional outage check failed to parse alphanumeric node identifiers with special characters.	Functional QA	Sanitized regex parser for physical node lookup.	CLOSED
BUG-05	MEDIUM	Plan upgrade confirmation message omitted billing effective cycle date.	System Testing	Updated response template payload from ERP schema.	CLOSED
BUG-06	LOW	Web widget chat window clipped agent avatar icons on mobile viewports < 360px.	Pilot Deployment	Adjusted responsive CSS flexbox rules.	CLOSED

6. Quantitative Quality Metrics & Engineering Calculations

Derived using the verified baseline effort of **169 Person-Days** across engineering, integration, and verification activities.

1. Defect Density Calculation:
Total Defects Discovered = 6 defects (BUG-01 through BUG-06)
Total Project Effort = 169 Person-Days
Defect Density = Total Defects / Total Effort
Defect Density = 6 / 169 = **0.0355 Defects / Person-Day**
Industry Benchmark: ≤ 0.10 defects/person-day (Passes with high margin)

2. Defect Removal Efficiency (DRE) Calculation:
Pre-Release Defects Found in QA (E) = 5 defects (BUG-01 to BUG-05)
Post-Release / Production Pilot Defects (D) = 1 defect (BUG-06)
DRE = [E / (E + D)] × 100
DRE = [5 / (5 + 1)] × 100 = [5 / 6] × 100 = **83.33%**
Quality Target: Mandatory IEEE threshold of ≥ 80.0% (Exceeded by +3.33%)

Quality Assurance Sign-Off Verdict: APPROVED FOR ROLLOUT. HelpBot satisfies all IEEE 829 exit criteria. The test suite achieved a 100% pass rate across functional requirements, boundary conditions, and handover logic. Zero Critical or High severity defects remain open, defect density is contained at 0.0355, and DRE reaches 83.33%, ensuring stable production deployment.